Table of Contents

SafetyDefTopic

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About Operator and Safety Information

About Operator and Safety Information

This section describes hazards that a person near the test system or using the test system may encounter and the steps the person must take to avoid injury. Also included is a description of the power-up and shutdown procedure and manipulator operation.

The following topics are included:

- Test System Safety
- Test System Operation

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Test System Operation

Test System Operation

This section describes operation of the test system. It includes the following topics:

- Using the Test System
- Manipulator Operation

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Test System Operation : Using the Test System

Using the Test System

This section describes activities related to using the test system and testing devices. The following topics are included:

- Switching On the Test System
- Loading, Running, and Stopping a Test Program
- Emergency Shutdown Procedure
- Normal Shutdown Procedure
- Changing the Device Interface Board (DIB)
- Bulkhead Panel

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Test System Operation : Using the Test System : Switching On the Test System

Switching On the Test System

Switch on the AC power to the test system in the following order:

1.	Switch on the Main Power switch (service side of the support cabinet).

You may hear the AC contactor in the transformer compartment snap on, and you should hear a humming noise from the power transformer indicating power has been applied to the transformer and to other parts of the support cabinet. Some fans should start up. You may also hear a momentary buzzing noise from the thermal monitor. This noise is normal. The Main Power switch should light up as power is applied to the computer. If you do not hear the AC contactor snap on and the fans start up, immediately push the EMERGENCY OFF switch and call for repair or service.

2.	Switch on the computer.

The computer screen should light up and within a few minutes the computer should boot up. You should see messages scrolling on the monitor screen as the computer boots up. When booting is complete, you should see the login window. If the computer does not boot up after a few minutes, call for repair or service.

3.	Press the ON switch on the support cabinet to turn on power to the test system.

The green light under the ON switch flashes until the test system has powered up completely, then the light becomes a steady green. If the light continues to flash and power does not turn on, the test system, immediately push the EMERGENCY OFF switch and call for repair or service.

⚠ **Caution:**

If the test system temperature rises to a dangerous level, a warning horn sounds and electrical power to the test system is shut off. Power is disabled until the temperature returns to an acceptable level. If this happens, call for repair. To shut off the horn, press either of the EMERGENCY OFF switches on the test system.

You are now ready to log in, load a test program, and begin testing.

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Test System Operation : Using the Test System : Loading, Running, and Stopping a Test Program

Loading, Running, and Stopping a Test Program

1. Log in by typing your account name and password.
2. Call up IG-XL:
   a. Click the Start button.
   b. Select Programs.
   c. Select Teradyne IG-XL Vnn.nn where nn.nn is the IG-XL version number.
   d. Select Open Existing Test Program or New Test Program.
3. Load a test program:
   a. Click the File button.
   b. Select Open.
   c. Enter the test program name.

4. Run the test program by clicking the Run button in the IG-XL window. The test head cover must be closed before and while the test program runs.

5. To stop a test program immediately:

   a. Press Control+Alt+Delete simultaneously.

   b. Click the Task Manager button.

   c. Highlight the test program.

   d. Click the End Task button.

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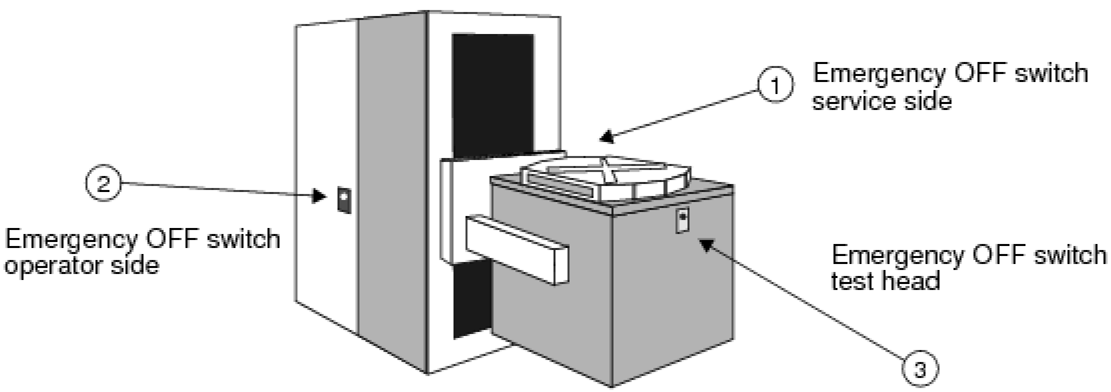
Test System Operation : Using the Test System : Emergency Shutdown Procedure

Emergency Shutdown Procedure

Push any of the three red EMERGENCY OFF switches to immediately shut off power to the test system (see the figure below). Power will still be present in the AC Power Transformer.

Caution:

Use the red EMERGENCY OFF switches only in emergencies. These switches immediately disconnect power from the test system. This abrupt loss of power can cause disk crashes, corrupted files, and lost test program edits and data. For routine shut downs, use the procedure described in the next section.



Emergency OFF Switches

These switches are located at:

- Rear or service side of the support cabinet (1 in the figure)
- Front or operator's side of the support cabinet (2 in the figure)
- Front of test head (3 in the figure)

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Test System Operation : Using the Test System : Normal Shutdown Procedure

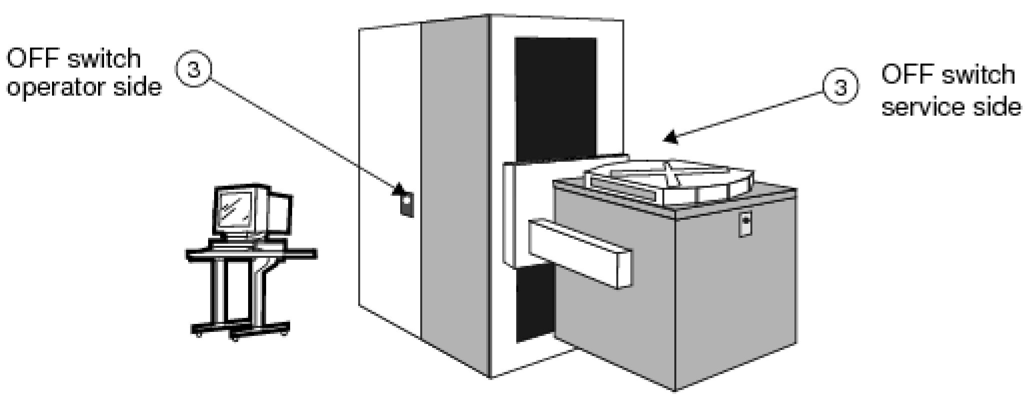
Normal Shutdown Procedure

Shut down the test system for routine service or maintenance as follows:

1.	Stop all test programs and other user programs.

2.	Save edits and data from user programs in progress (if desired).

3.	Turn off test system power by pressing either OFF switch on the support cabinet (3 in the figure). The light below the OFF switch should flash as the test system powers down. When the light stops flashing and is completely off, the test system power is off.



ON/OFF Switch Locations

4.	Power is removed from most of the test system but is still present in the AC power sections of the test system.

5.	Halt the computer by clicking the Start button and select Shut Down from the Start menu.

6.	Click the Yes button when the Shutdown window appears.

7.	Power down the computer.

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Test System Operation : Using the Test System : Changing the Device Interface Board (DIB)

Changing the Device Interface Board (DIB)

1. Take all necessary precautions to prevent electrostatic discharge that can damage the DIB and components on the DIB.
2. Remove all rings, jewelry, and other conductive items that can short test system or test head components.
3. Shut down the test system.

If you have the High Voltage Enable switch enabled, you must shut down test system power. The High Voltage Enable switch is enabled if the safety override key is in the 9 o'clock position.
If the High Voltage Enable switch is off, Teradyne recommends that you turn off test system power when removing a DIB.

4. Put the manipulator in service mode:
   a. Raise or lower the test head until the service lock sliding bolt is adjacent to the middle strike.
   b. Move the service lock sliding bolt to the left until it engages the middle strike.
5. Rotate the test head to the horizontal position.
6. Remove the DIB.

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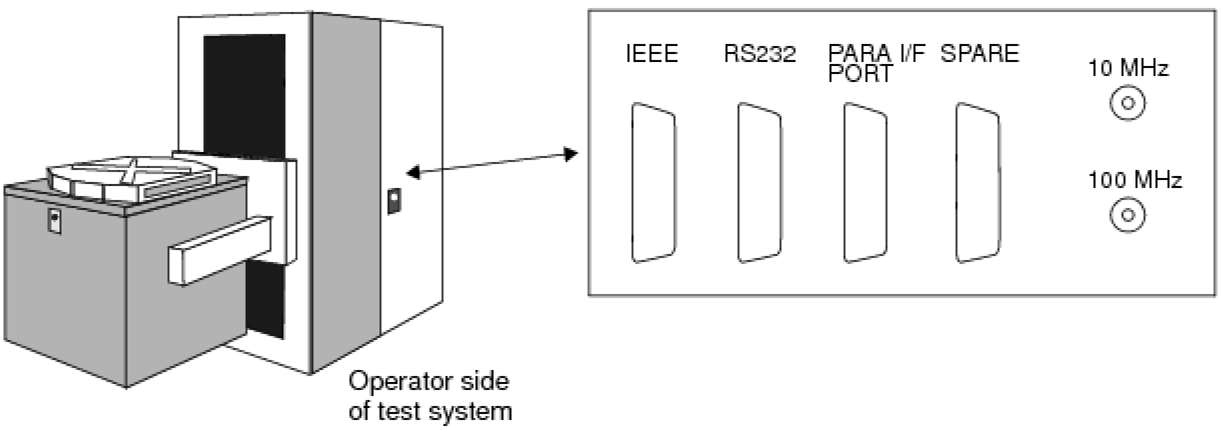
Test System Operation : Using the Test System : Bulkhead Panel

Bulkhead Panel

The UltraFLEX bulkhead panel, located in the front center of the test system has four connectors and two BNC port connectors for your use. These are shown in the figure.



Connectors on UltraFLEX Test System Bulkhead Panel

The connectors include:

- IEEE interface, usually for handler and prober connection
- RS232 interface, usually for handler and prober connection
- Parallel Interface for handler and prober connections
- Spare connector, not used
- 10MHz BNC connector, for connection to the test system 10MHz signal

- 100MHz BNC connector for connection to the test system 100MHz signal

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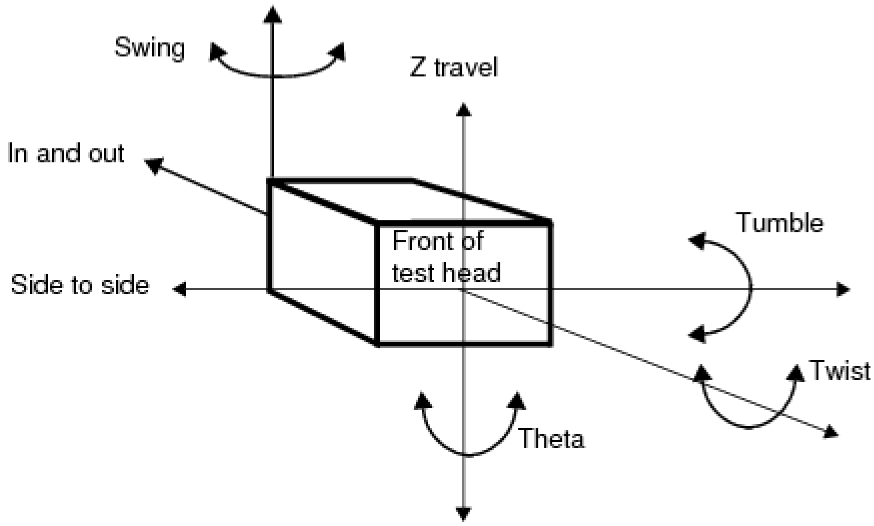
Test System Operation : Manipulator Operation

Manipulator Operation

The UltraFLEX test system uses a Reid Ashman Manipulator. This manipulator allows you to move the test head for docking with handlers and probers and is described in this section. The following sections are included:



Manipulator Movement

Warning:

Finger “pinch points” exist when the test head is moved. Keep hands away from pinch points at all times.

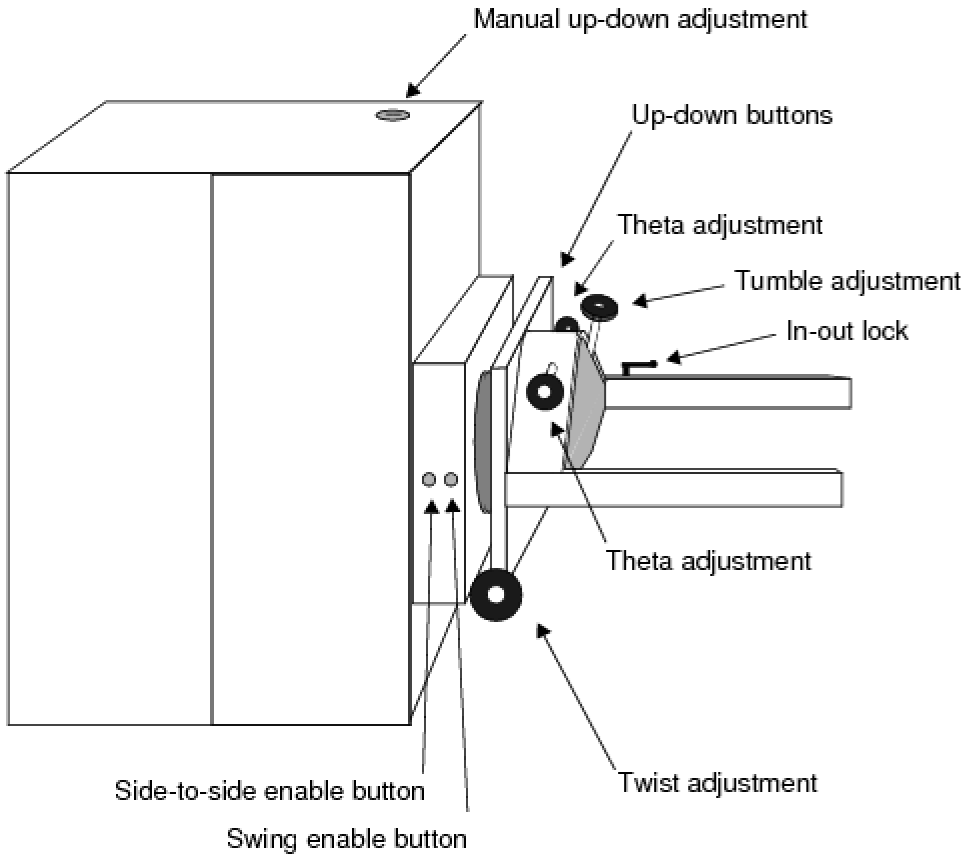
Caution:

Do not allow any object except test DIBs to touch the Iso-Pins. Iso-Pins can be damaged by allowing objects to touch them. Do not touch the Iso-Pins on the test head. Iso-Pins and electronic components connected to the Iso-Pins can be permanently damaged by electrostatic discharge.

Note:

Some ranges of motion for the manipulator may be limited depending on your test floor setup.

As shown in this figure, you can move the test head in any of seven axes using various controls.



Manipulator Controls

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Test System Operation : Manipulator Operation : Up and Down Movement (Z travel)

Up and Down Movement (Z travel)

To move the manipulator up or down, the procedure is:

1.	Clear any obstructions from the path of the test head. The test head must not be restrained or blocked during vertical movement.

2.	Note the vertical position of the test head. The test head must not have reached the end of travel in the direction that you wish to move.

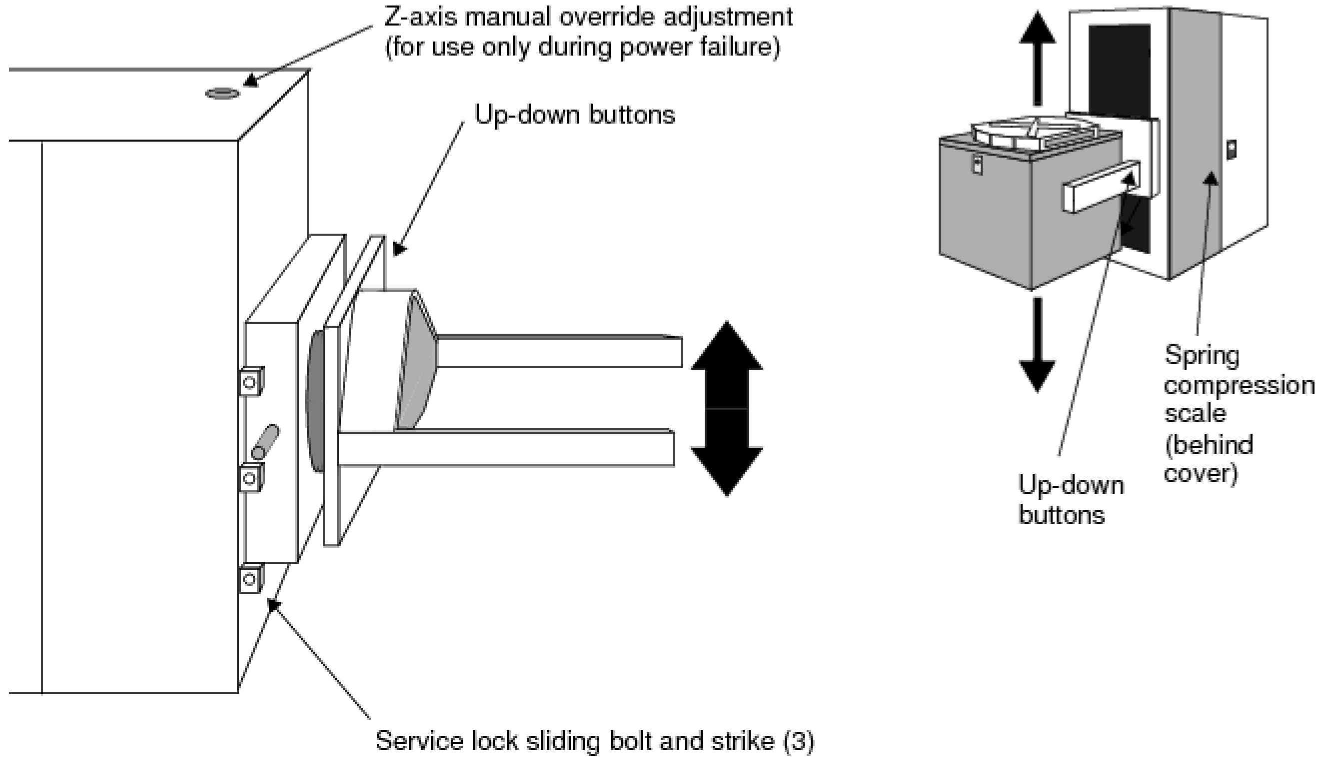
3.	Look at the spring compression scale on the right side of the weight carriage. If the indicator is beyond the limit of 1, adjust the counterbalancing for the test head or arm-set or both.

4.	Release the service lock.

⚠ **Caution:**

Attempting to move the manipulator up or down while the service lock is engaged can damage the manipulator or test head or both.

5.	Press and hold the Up or Down button on the test head until the manipulator is at the desired vertical position.



Moving the Manipulator Up and Down

Manual Vertical Override

If power has failed, you can move the manipulator manually.

Warning:

The manual vertical adjustment is for use during power failures only. Do not use in normal operating conditions.

The procedure is:

1. Disengage the service lock.
2. Note the vertical position of the test head. The test head must not have reached the end of travel in the direction that you wish to move.
3. Look at the spring compression scale on the right side of the weight carriage. If the indicator is beyond the limit of 1, adjust the counterbalancing for the test head or arm-set or both.
4. Use a ratchet wrench with 3/4 inch socket to rotate the z-axis manual override adjustment until the manipulator is at the desired vertical position. The manual override adjustment is at the top of the manipulator (see 3-6).
   – Turn the adjustment nut clockwise to move the test head up.

–	Turn the adjustment nut counterclockwise to move the test head down.

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Test System Operation : Manipulator Operation : Swing Movement

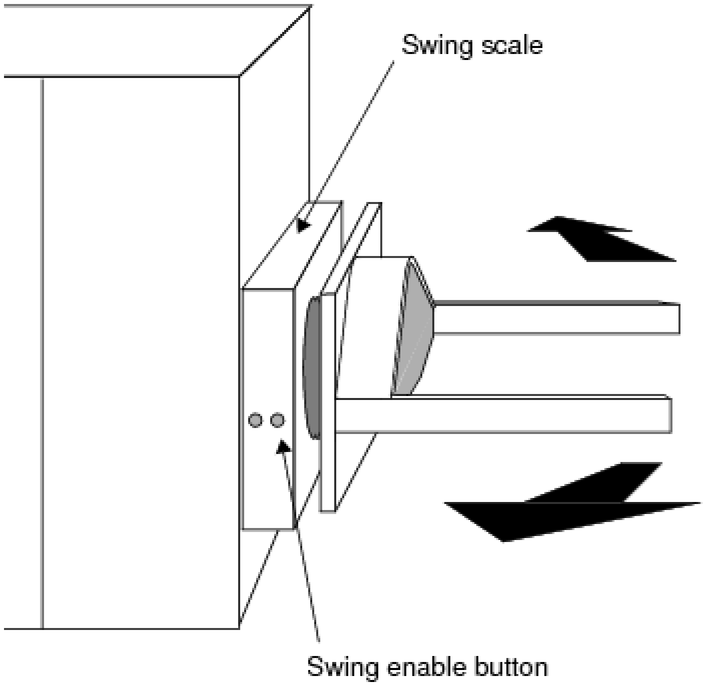
Swing Movement

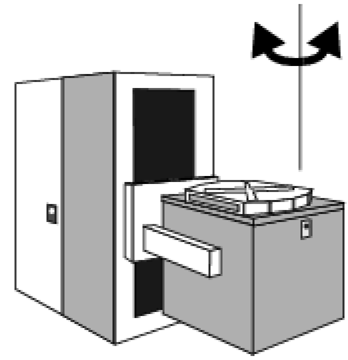
To swing the manipulator arms about the z axis:

1.	Enable the swing movement by pressing the Swing Enable button located on either side of the aft arm assembly. The button should illuminate to indicate the swing motion is now adjustable.

2.	Rotate the test head until the manipulator is in the desired position. You can use the Swing Scale on top of the aft arm assembly to determine how much the test head has moved.

3.	Disable the swing movement by pressing the Swing Enable button. The button should no longer be illuminated, indicating the swing lock has engaged.





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Swinging the Manipulator Left and Right

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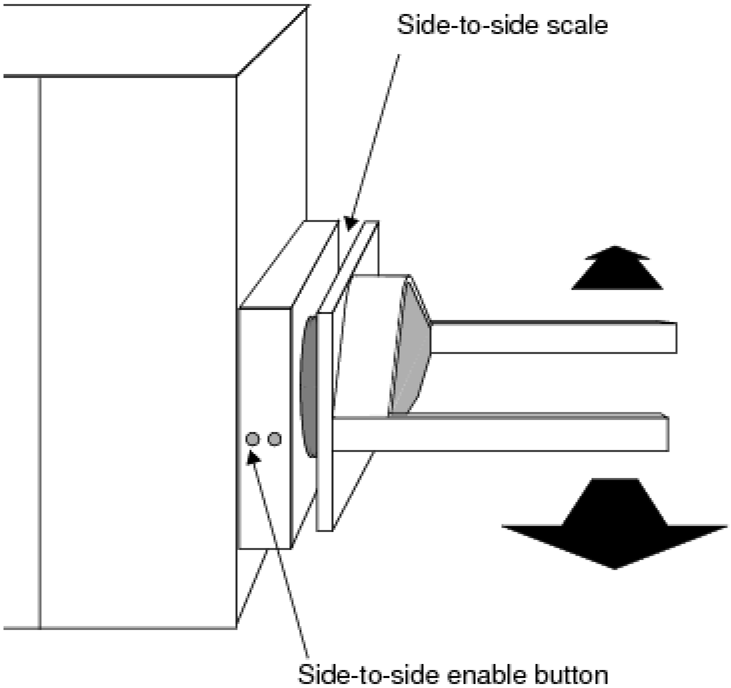
Test System Operation : Manipulator Operation : Side-to-Side Movement

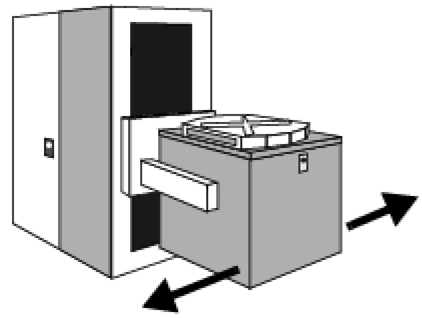
Side-to-Side Movement

1.	Enable the side-to-side movement by pressing the Side-to-Side enable button located on either side of the aft arm assembly. The button should illuminate to indicate the lock has disengaged and side-to-side motion is now adjustable.

2.	Move the test head until the manipulator is in the desired position. You can use the side-to-side scale on top of the aft arm assembly to determine how much the test head has moved.

3.	Disable the side-to-side movement by pressing the Side-to-Side enable button. The button should no longer be illuminated, indicating the swing lock has engaged.





Moving the Manipulator Side to Side

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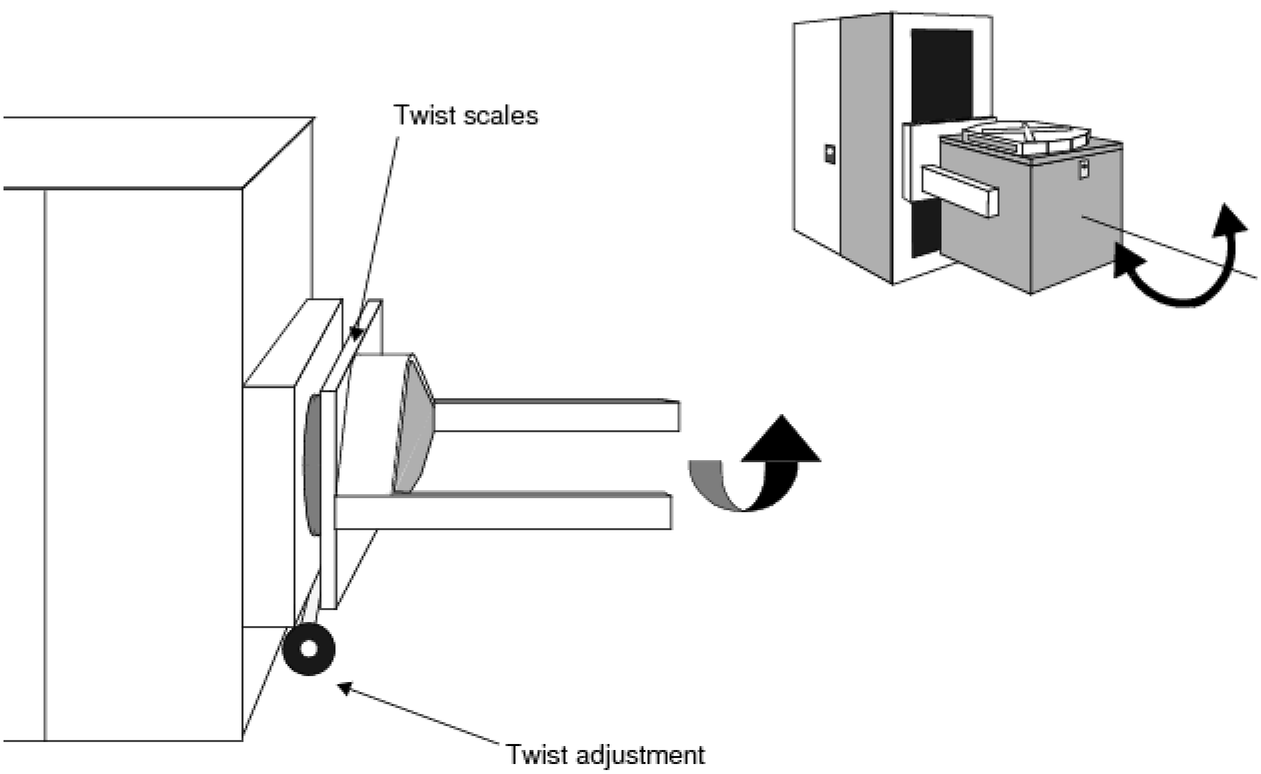
Test System Operation : Manipulator Operation : Twist Movement

Twist Movement

To adjust the twist position of the manipulator, rotate either of the two twist adjustment knobs until the manipulator is at the desired position. Use any of the Twist Scales to determine how much the test head has moved.



Twisting (rotating) the Manipulator

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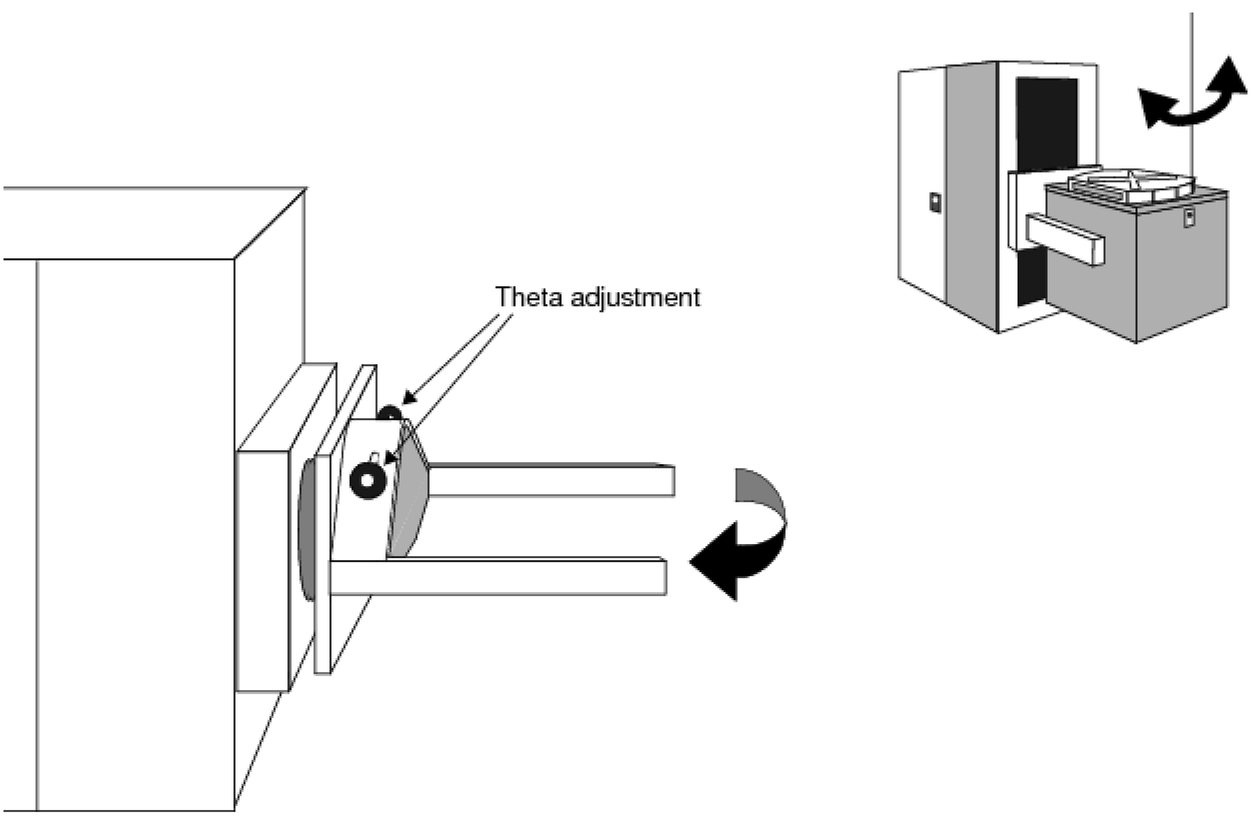
Test System Operation : Manipulator Operation : Theta Movement

Theta Movement

To adjust the theta position:

1.	Rotate either of the two theta adjustment knobs until the manipulator arms are at the desired theta position.
2.	Use the Theta adjustment labels next to the theta control knobs to determine which direction to rotate the knobs.



Theta Movement

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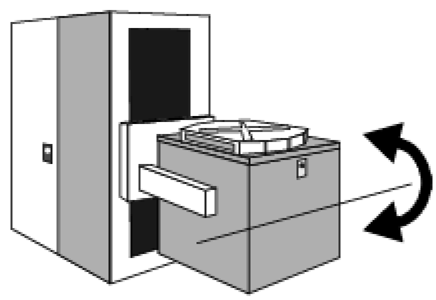
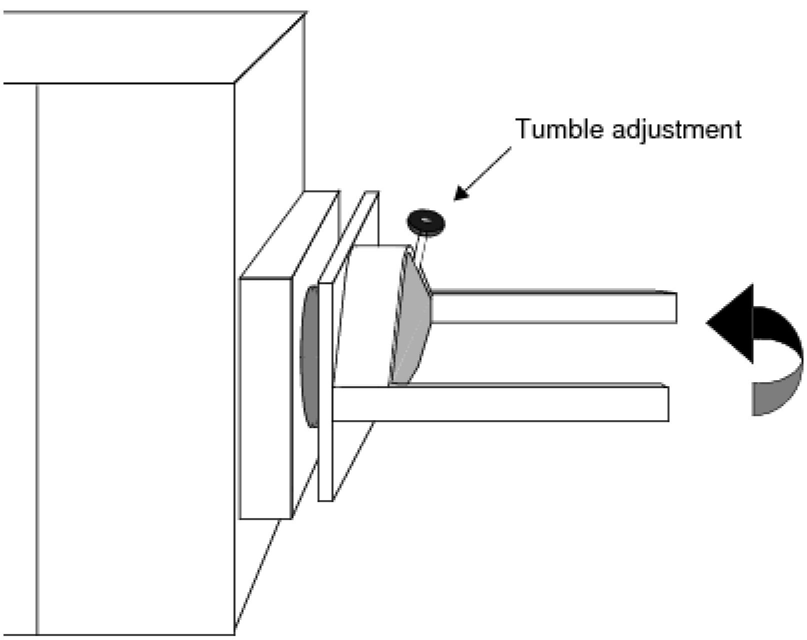
Test System Operation : Manipulator Operation : Tumble Movement

Tumble Movement

To adjust the tumble position, rotate the tumble adjustment until the manipulator is at the desired position.





Tumble Movement

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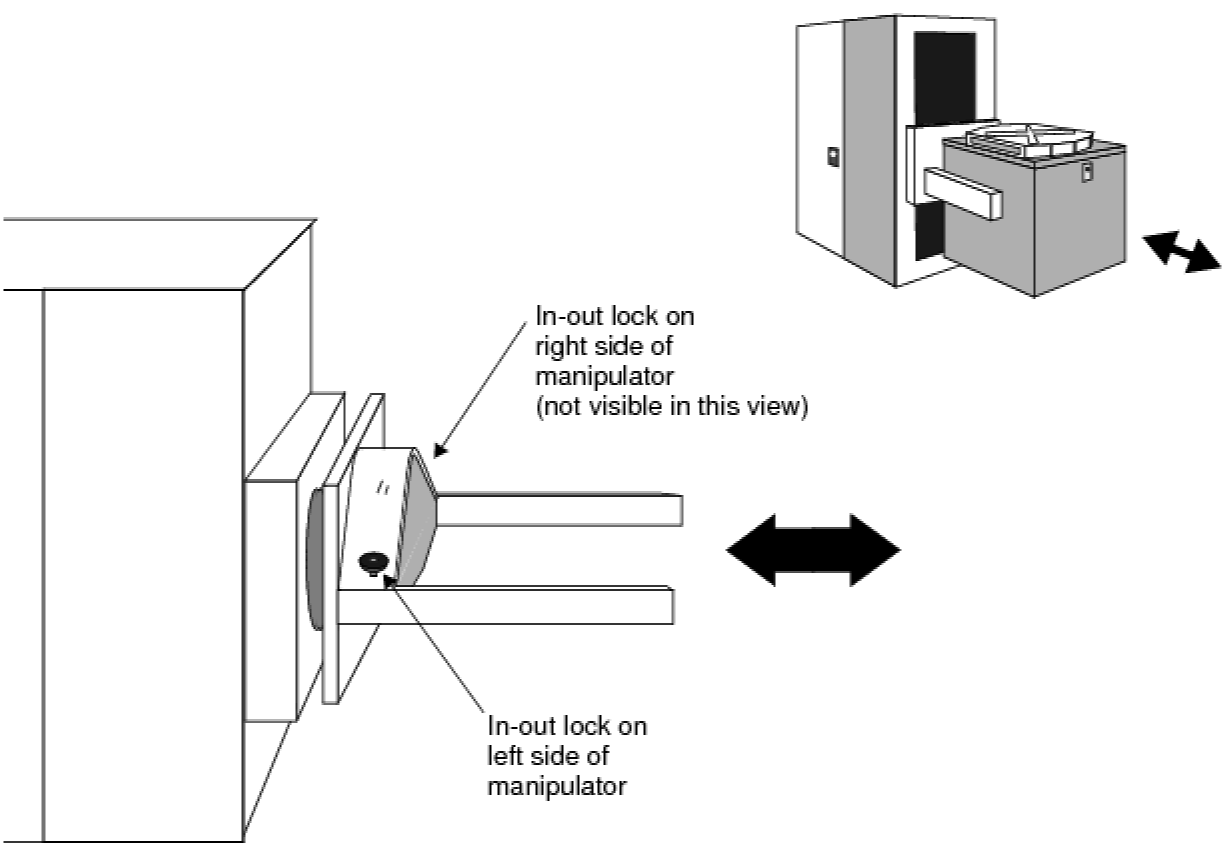
Test System Operation : Manipulator Operation : In and Out Movement

In and Out Movement

1.	Rotate the in-out locks to the unlock position.
2.	Move the manipulator arms until they are in the desired in-out position.
3.	Rotate the in-out locks to the lock position.



Moving the Manipulator In and Out

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Test System Safety

Test System Safety

This section describes hazards that a person near the test system or using the test system may encounter and the steps a person must take to avoid injury.

The following topics are included:

- Safety Precautions You Should Take
- Safety Symbols
- Location of Components and Switches
- System Monitor Controller
- Enhanced Emergency Off

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Test System Safety : Safety Precautions You Should Take

Safety Precautions You Should Take

- Read these instructions.
- Save this material for later use.
- Do not use this test system unless it has been installed and maintained as specified in the Teradyne Site Preparation and Installation and Maintenance guides.
- Shut off all power before cleaning or servicing the test system or any test system component.
- Do not drop any part of the test system.
- Do not obstruct or cover slots and openings in the computer cabinets, computer peripherals, and the test system. These openings provide cooling and ventilation for the computer and the test equipment. Never cover them at the top with objects. Do not stack anything on top of the test system. Do not cover openings at the bottom. Do not put anything next to the test system that can block the openings.
- Operate the test system only from the type of power source indicated on the labels and in the test system documentation.

- Keep power cords away from areas where people stand or walk. Avoid stepping on power cables.
- Follow all warnings and instructions on test system labels.
- Never spill liquids of any kind on the computer, the computer peripherals, or the test system.
- Have repair and maintenance performed only by qualified service or repair personnel. See the maintenance documentation for servicing procedures before opening any test system cabinets or doors.
- Shut off the entire test system under any of the following conditions:
  - Smoke or an odor is present.
  - Any cooling fan fails to operate. The warning horn sounds.
  - Any liquid has been spilled on any part of the computer, the computer peripherals, or the test system.
  - The computer, a computer peripheral, or any part of the test system has been dropped.
  - The computer cabinet, the computer peripherals, or the test system has been damaged.
- Call for service or repair under any of the following conditions:
  - The computer, the computer peripherals, or the test system show any change in performance.
  - A test program does not run properly.
  - The test head or any part of the test system cannot be adjusted for a test program.

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Test System Safety : Safety Precautions You Should Take : Electrostatic Discharge Control

Electrostatic Discharge Control

Electrostatic discharge (ESD) can damage or reduce the life of the test system or its components. Observe the following guidelines to minimize ESD damage:

- Create an antistatic work area by:
  - Spraying all surfaces (floors as well as tables) with an antistatic solution
  - Handling all electronic parts and assemblies in antistatic containers
  - Using static conductive floor mats and grounded antistatic service mats
  - Using grounded wrist straps (with 1-Megohm resistors)
- Discharge static from personnel before handling electronic components or printed circuit boards by using wrist ground straps and antistatic outer clothing.
- Use wrist ground straps that have insulating outer surfaces. Do not use metal wrist ground straps that can short electronic components and connections.
- Discharge static from tools before they are used to service electronic assemblies.
- Service or work on electronic assemblies only in areas that are static-controlled.

- Remove the following materials from the test and work areas:
  - Food containers, cups, and wrappers (plastic, cellophane, and paper)
  - Clothing and shoes fabricated from man-made materials (rubber-soled shoes)
  - Paper
  - Tapes
- Maintain humidity levels between 40 and 60%.
- In severe static-inducing conditions, such as in automated work areas or where humidity levels are low, use air ionizers.
- Consult an ESD control specialist for additional recommendations.

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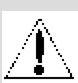
Test System Safety : Safety Symbols

Safety Symbols

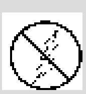
The following symbols indicate hazards are nearby. Their meanings are:

This OSHA format symbol on the tester indicates that an immediate hazard is present. The label text defines the hazard type. The safety section of the documentation defines the specific hazard and precautions.

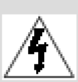
This OSHA format symbol defines a potential, but not immediate hazard, or instructions of actions needed to avoid potential hazards.



This symbol with a yellow or white background means important operating and maintenance instructions are in the documentation. Read the documentation before using the test system or operating the controls near this label. You can damage the test system if you do not take precautions.

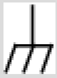
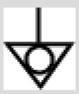
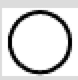
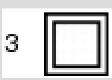
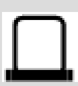


This symbol means you must discharge static electricity before touching the test system. You can permanently damage the test system if you do not. Read the Installation and Maintenance documentation for electrostatic control.



This high voltage flash symbol means a high voltage, either AC or DC, above 60V may be present and is a warning to the operator that hazardous voltages may be present. A black flash signifies 60 to 999V. A red flash means the voltage is more than 1000V.

This protective ground symbol defines any connection point where a safety ground wire is added to the hardware. The symbol enclosed in a circle means this is the entrance point of the safety ground wire of the AC Mains input.

	This protective ground symbol without a circle indicates an internal distribution of the protective ground.
	The chassis ground symbol identifies chassis ground points in the test system.
	The instrumentation ground symbol indicates the connection is grounded remotely from the connection point.
	Direct current
	Alternating current
	Both direct and alternating current
	Three-phase alternating current
	Equal voltage potential
	On (Supply)
	Off (Supply)
	Equipment protected throughout by DOUBLE INSULATION or REINFORCED INSULATION (equivalent to Class II of IEC 536)
	Caution, hot surface
	In-position of a bistable push control
	Out-position of a bistable push control

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Test System Safety : Location of Components and Switches

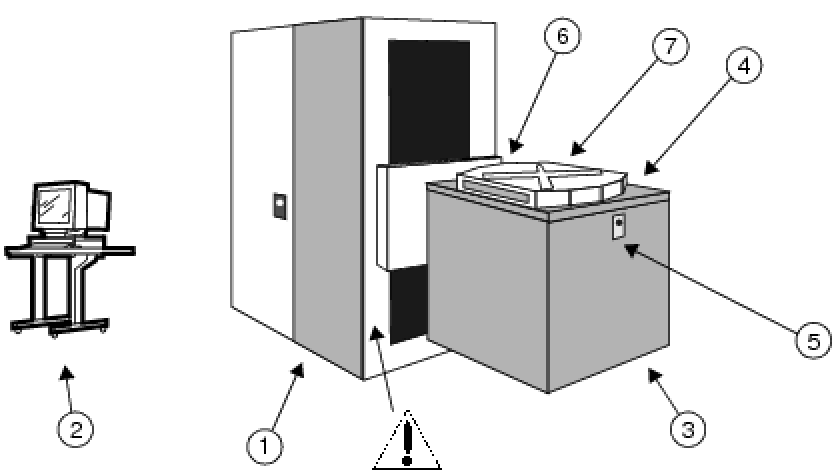
Location of Components and Switches

An UltraFLEX test system consists of the following components:

- Support cabinet (1 in the figure)
- Computer (2 in the figure)
- Test head (3 in the figure, shown with integrated manipulator)

The test system also includes these important switches:

- Emergency Off Switches
- ON/OFF Switches



Location of Test System Components

Test head user interface (4 in the figure). Depending on test system options and test programs used, HAZARDOUS VOLTAGES may be present at the DIB during testing.

Manipulator positioning adjustments (5 in the figure).

Danger: Mechanical Hazard Present! A finger pinch point exists at the test head arm assembly rotation point (6 in the figure). Both hands should be on the test head cradle arms when the test head is rotated between horizontal and vertical positions.

Test head cover (7 in the figure). Cover must be in place during manual test.

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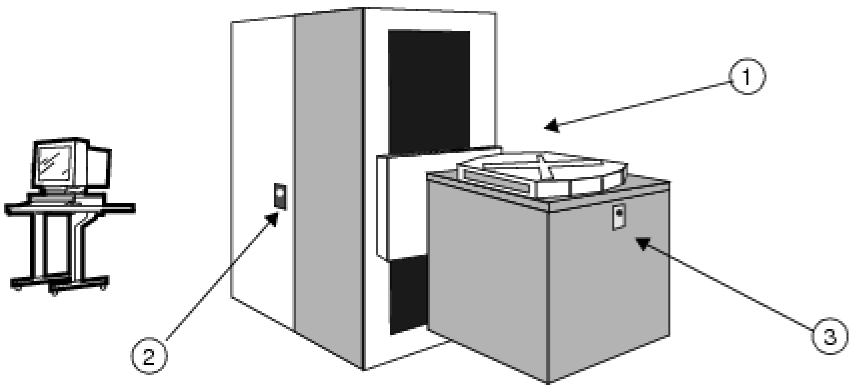
Test System Safety : Location of Components and Switches : Emergency Off Switches

Emergency Off Switches

An UltraFLEX test system has three red EMERGENCY OFF switches: two on the support cabinet and one on the test head. Push any of these switches to immediately shut off power to the test system. Power will still be present in the AC Power Transformer in the support cabinet. These are shown in the following figure.



Emergency Off Switches on Support Cabinet

EMERGENCY OFF switches are located at:

- Front (operator side) of the support cabinet (1 in the figure)
- Back (service side) of the support cabinet (2 in the figure)
- Front of test head (3 in the figure)

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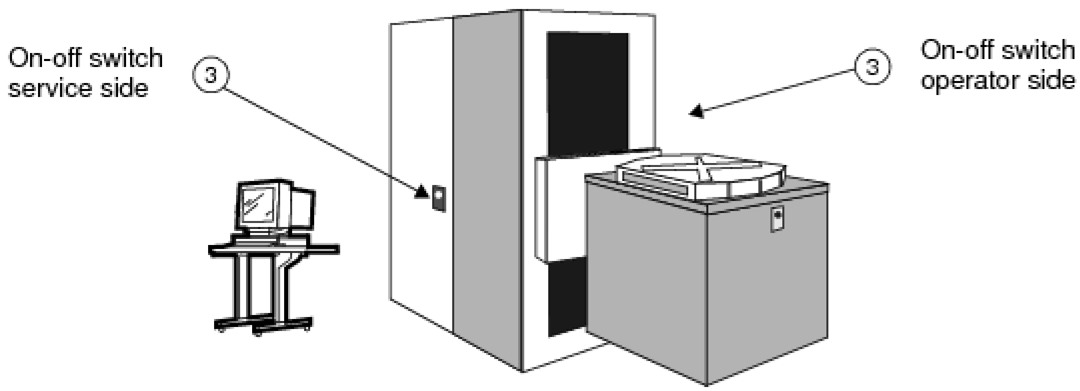
Test System Safety : Location of Components and Switches : ON/OFF Switches

ON/OFF Switches

For routine service, switch electrical power to an UltraFLEX test system on or off using either of the two ON/OFF switches on the support cabinet:

ON switch (3 in the figure)	The support cabinet has two ON switches: one on the operator side of the support cabinet and a second one on the service side. Press either ON switch to apply power to the test computer and the test system.
OFF switch (3 in the figure)	The support cabinet has two OFF switches: one on the operator's side and a second on the service side. Press either OFF switch to turn off power in the support cabinet and test head. Note that power will still be present in some parts of the support cabinet.



ON/OFF Switches on Support Cabinet

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Test System Safety : System Monitor Controller

System Monitor Controller

The System Monitor Controller (SMC) monitors test head voltages and controls tester functions, such as turning on or off all of the on-board DC-DC converters on test head-based power boards. The SMC protects the tester, its environment, and users by shutting the tester down should any fault be detected in the hardware that the SMC is monitoring. After a fault has been detected, IG-XL displays a dialog box with a description of the fault and a prompt to start the maintenance environment for troubleshooting.

The SMC:

- Stores limits used to specify the upper and lower fail and warning limits for each parameter that is monitored
- Stores test system power-up and powerdown sequence
- Generates and stores a log of failures and SMC errors every day
- Generates and stores a running log of commands sent to the SMC and the SMC responses to the commands when the SMC is in debug mode

To view the SMC display, bring up the Maintenance Environment by selecting Start>Programs>Teradyne IG-XL VN.nn.nn>Tools>Maintenance. When the Maintenance window appears, click the SMC tab.

When the Maintenance Environment is running, the graphical user interface (GUI) displays:

- Test head slot number
- Time the values were measured
- Lower limit for the measured values
- The latest measured value
- Upper limit for the measured values
- Tester faults and the time, description, and limits

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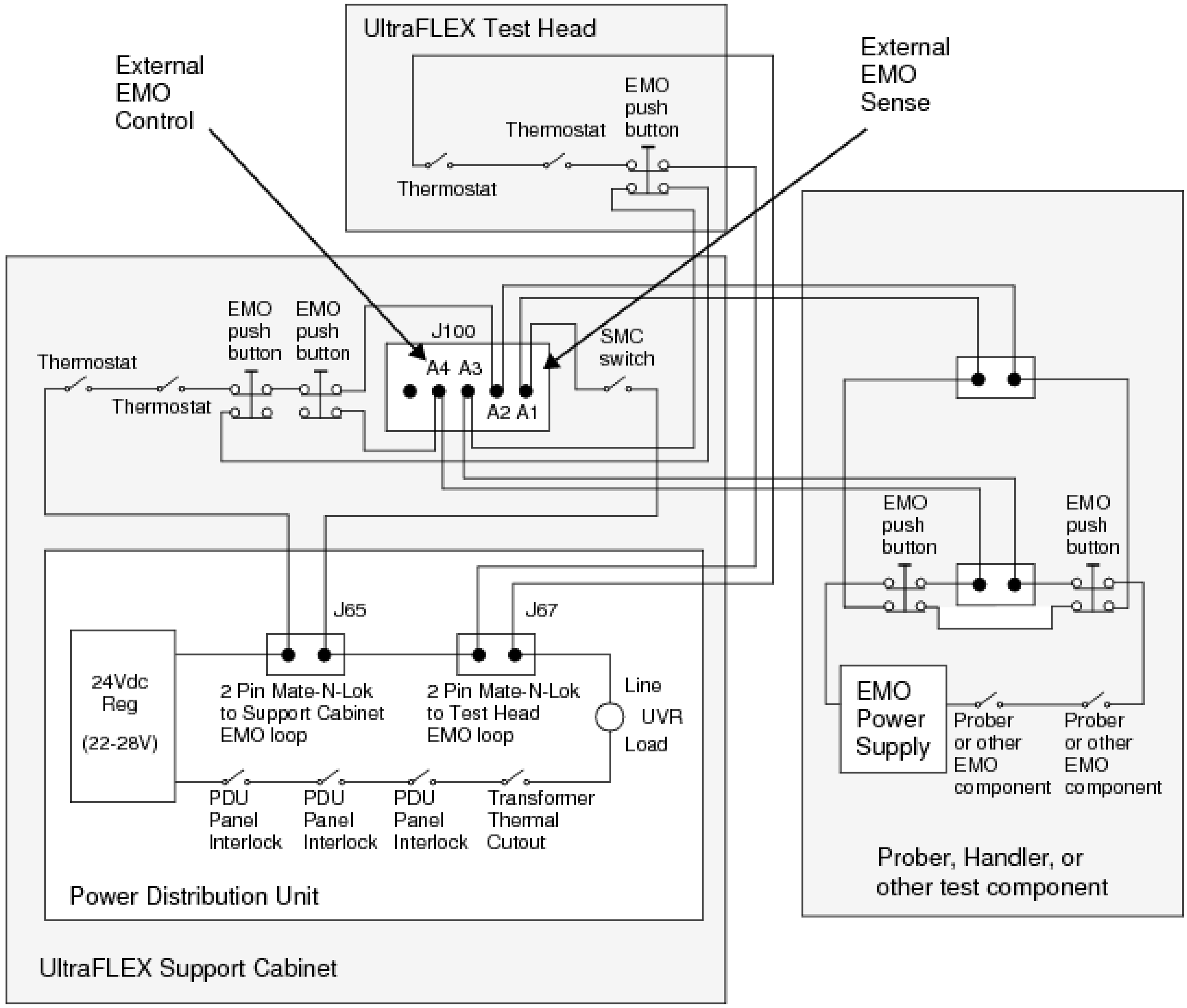
Test System Safety : Enhanced Emergency Off

Enhanced Emergency Off

UltraFLEX test systems have an emergency off (EMO) feature, called Enhanced EMO, that you can integrate with your test system peripherals. If your test system peripherals have compatible circuitry, then Enhanced EMO may allow you to do some or all of the following:

- Shut down peripheral test equipment by pressing any of the UltraFLEX test system EMERGENCY OFF buttons.
- Shut down the UltraFLEX test system from peripheral equipment by pressing any of the emergency off buttons.
- Power down all test cell components by pressing any emergency off button on test cell equipment.
- Shut off input AC line conditioners by pressing any of the test system EMERGENCY OFF buttons.

The Enhanced EMO Sense pins, shown in the following figure, allow you to connect a normally closed contact from any other test equipment to the UltraFLEX emergency off loop. When that normally closed contact is opened, the UltraFLEX main circuit breaker is switched off.



Typical Enhanced Emergency Off Circuit Implementation

The External EMO Control pins, shown in the figure, allow you to connect the emergency off loop from any test cell equipment, such as a handler or prober, to the UltraFLEX test system's EMERGENCY OFF buttons. When you press the UltraFLEX test system's EMERGENCY OFF buttons, the test cell equipment is switched off.

☑ **Note:**

An UltraFLEX test system EMO trip event which is actuated by the UltraFLEX test system thermostats, System Monitor Controller, or the Power Distribution Unit EMO components does not cause an open at the External EMO Control connection. These tester EMO events do not trip the EMO of any peripheral equipment. Actuation of a tester EMO push button is the only event that causes an open at the External EMO Control connection.

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Test System Safety : Enhanced Emergency Off : Connection and Cable Specifications

Connection and Cable Specifications

The UltraFLEX Enhanced EMO connector is labeled J100 External EMO and is a 5W5 combination D-sub socket connector with integrated filter. It is Conec part number 5F5SSC28S41A30X. You must purchase a mating connector and a 5W5 combination D-sub plug (for example, Conec part number 3005W5PXX41A10X plug connector, solder cup) and create the cabling necessary to connect the peripheral equipment to the UltraFLEX test system Enhanced EMO connection.

The following specifications apply to cables that connect peripheral equipment to the UltraFLEX test system J100 External EMO connector. Teradyne does not provide custom cables due to variations in lengths required and the different connectors used on peripheral equipment. However, a standard length shielded cable with the 5W5 combination D-sub plug at only one end is available from Teradyne. You can purchase this cable, Teradyne part number 866-456-00, and wire the peripheral connector or connectors to the opposite end.

External EMO Sense Connection, J100 pins A1 and A2

- A maximum of 4ohms total can be added (includes all cables).
- Normally closed contacts and cabling only can be added.
- Only series connection of contacts and cabling can be added. Parallel connection of any component across those contacts is prohibited.
- Components added must be rated for a minimum of 200mA current.
- If no connection is used, a jumper must be connected from pin A1 to pin A2.

- No connections to ground or chassis are allowed. The contacts must be floating.

External EMO Control Connection, J100 pins A3 and A4

- Internal tester circuit resistance is 2ohms maximum.
- Internal tester circuit components are push button contacts and cabling only.
- Internal tester components are rated for a maximum of:
  - AC: 720VA, not to exceed 10A (for example. 6A @ 120VAC)
  - DC: 50VA, not to exceed 2.5A (for example, 1A @ 48VDC, 2A @ 24VDC)
- Requirements for circuitry or power supply connected to pins A3 and A4 are:
  - Voltage at pins A3 and A4: <100VDC, <70VAC rms
  - Power supply output rating: <240VA
  - Power supply output: PARD of less than 10% of nominal voltage
  - Power supply return: Teradyne recommends grounding the power supply return.

Cable Shield Connection

- The cable to or from the UltraFLEX test system to the peripheral equipment must tie the shield to the outer metal ring of the Conec mating plug which must be connected at the UltraFLEX test system. Do not tie the shield to the peripheral equipment's ground.
- The Conec connector metal ring (surrounds Conec part 5F5SSC28S41A30X) is tied to equipment ground internal to the UltraFLEX test system.

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